Unified Product Command Center

Feature plan, v2 · Vacademy super admin portal · 3 November 2026

> **The plan:** add a Command Center module to the Vacademy super admin portal. It brings website traffic and funnel events from the four product sites together with logins, usage, AI token use and platform billing from the shared backend, into one analytics store. From the same screens you can create demos, start trials, and suspend, terminate or recreate accounts. v2 builds on what the portal and backend already have instead of rebuilding it.

0. What changed from v1

- **Product names corrected:** Vacademy.io, Voomsales, Telleo and Evalezy.
- **Reuse, don't rebuild:** Razorpay billing, demo provisioning, login/activity data, token metering and change history already exist (§1.3).
- **No second Razorpay webhook.** Revenue reads admin-core's platform billing, so there is one source of truth for payments.
- **Where things run:** the portal has no server, so ingestion runs in community-service (§2.1).
- **External connections listed one by one,** with what exists and what is missing, checked against the code on 3 Nov 2026 (§2.2).
- **Sign-in entry points:** Vacademy.io, Telleo and Evalezy share `dash.vacademy.io`; Voomsales signs in at `app.voomsales.com` (§2.3).
- **New prerequisite:** a product entitlements table. Today an institute can carry only one product (§2.2, §6).
- **GA4:** `user_id` is now the real user ID, with `institute_id` as a user property (§3).
- **Consent and retention** section added for the visitor cookie (§3.4).
- **Funnel corrected:** login clicks come from existing customers, so they're out of the funnel. Demo and trial are two branches (§4).
- **State machine fixed:** a new Expired state, plus missing edges such as Suspended → Active and Demo → Active (§5).
- **Actions split:** per-product actions vs account-level actions (§5).
- **Destructive actions** are limited to named usernames (§1.2).
- **Metric definitions tightened:** visitors, churn, trial-to-paid and MRR (§7).
- **Rollout reordered:** Revenue moves to Phase 1 because its data already exists (§8).

1. Overview

Today each product site reports to its own analytics, usage sits in the backend, and billing sits in admin-core. Answering "*how many visitors became paying institutes this month, and on which product?*" means opening several tools and stitching the numbers together by hand.

Product	What it is	Website	Sign-in	Data the Command Center needs
Vacademy.io	LMS; base product, hosts the super admin portal	vacademy.io	dash.vacademy.io	Traffic, funnel clicks, logins, LMS usage, tokens
Voomsales	Sales CRM	voomsales.com	app.voomsales.com/login	Traffic, funnel clicks, logins, usage
Telleo	AI calling (also sold as an API at api.telleo.ai)	Own site	dash.vacademy.io	Traffic, funnel clicks, calls, tokens
Evalezy	Answer-sheet evaluation (also sold as an API at api.evalezy.com)	Own site	dash.vacademy.io	Traffic, funnel clicks, evaluations, tokens

`product` values used everywhere: `vacademy`, `voomsales`, `telleo`, `evalezy`. Vimotion is out of scope.

1.1 Decisions already made

- **One institute ID** is shared across all four products. The account list and Account 360 key on it. Voomsales will move to Vacademy auth-service and the same institute IDs (§2.3).
- **Token usage is already tracked.** The Command Center reads it and doesn't rebuild it.
- **Razorpay** is the payment gateway, through the platform billing that already exists in admin-core.

1.2 Who can use it

The PM is the main user in v1, and there is no full role model. But the portal already admits every username on the `PORTAL_ALLOWED_USERS` allowlist, which is enforced by the portal's Cloudflare proxy. So **Suspend, Terminate and Recreate** are also limited to a short list of named usernames. Every destructive action needs typed confirmation and writes an audit entry.

1.3 Already built: reuse, don't rebuild

Capability	Where it lives today	How the Command Center uses it
Platform billing via Razorpay: charges, payment links, mandates, mark paid, waive, refund, invoices, credit notes	admin-core platform billing; portal `plan-billing-api.ts`; HMAC-verified webhook `/payments/webhook/callback/razorpay-platform`; hourly reconcile jobs	Revenue screen and the Convert to paid action
Demo provisioning and expiry; login blocked when a demo expires	auth-service `/super-admin/v1/demo/provision`, `institute_trial` table, `TrialAccessGuard`; portal onboarding demo-accounts list	Create demo, Start / Extend trial
Logins, sessions, active users	`user_session`, `user_activity_log`, `daily_user_activity_summary`; auth-service `/active-users`, `/activity-trends`	DAU/WAU/MAU, last login, activation
AI token and credit usage	`ai_usage_event` (use case, source service, credits, real cost), `ai_token_usage`, credits tables	Tokens per institute, per product

Per-institute feature switches	Institute modules and presets	Suspend a single product without blocking login
Lead tags `PROD / LEAD / TEST / FREE_TRIAL`	Institute records	Mapped onto the new states, so there is one source of truth (§5)
Change history with reasons	Modules, pricing, plans, AI settings	Shown together in the Audit log screen

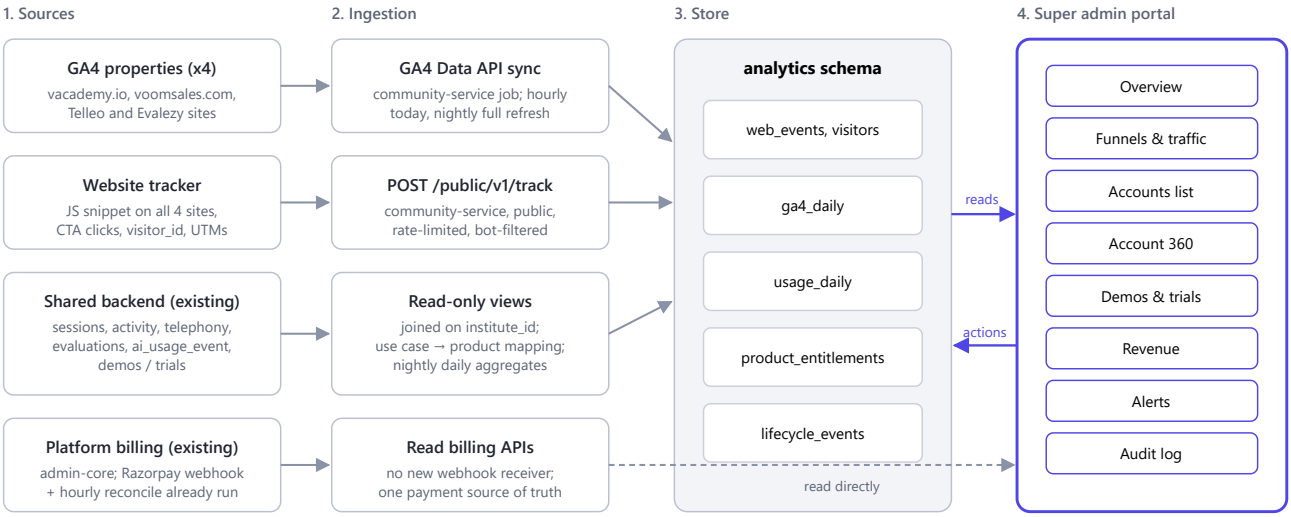
1.4 Out of scope for v1

- Rebuilding token metering or Razorpay integration.
- Access for sales or support teams (add roles later if needed).
- Customer-facing analytics for institutes, and Vimotion.

2. Architecture

The Command Center never queries GA4 or Razorpay live when a page opens. Website data is synced into one analytics store on a schedule, and backend data is read where it already lives. Dashboards load fast and every product is measured by the same definitions.

Sources feed one analytics store; the portal reads it and writes back lifecycle actions



Lifecycle actions call existing auth-service / admin-core endpoints and write lifecycle_events plus an audit entry.

2.1 Where ingestion runs

The portal is a static app behind a Cloudflare Pages proxy and has no server of its own. The GA4 sync job, the `/track` endpoint and the nightly aggregates run in **community-service**:

- It already hosts the public onboarding, pricing and status endpoints, allow-listed in `CommunityApplicationSecurityConfig`.
- It can use the `@Scheduled` + ShedLock pattern already used in admin-core.

Store: a separate `analytics` schema in the existing Postgres. Raw events move to ClickHouse or BigQuery only if volume makes dashboards slow.

2.2 External connections: what exists and what's missing

Checked against the backend and site repositories on 3 November 2026.

Connection	How	Exists today	Gap / owner
GA4 (4 properties)	GA4 Data API with one Google service account added as Viewer on each property	Voomsales GA4 `G-5VXCVMX3EH`; vacademy.io loads GA4 inside GTM `GTM-5C4DDJ6W`	**No GA4 code in the backend.** Telleo and Evalezy marketing sites aren't in any local repo, so owners and GA4 IDs are unknown. Need the property ID behind the GTM container.
Website tracker	Snippet on 4 sites posts to `/community-service/public/v1/track`	Public endpoint pattern only	New endpoint with rate limiting and bot filtering; snippet added to 4 site repos.
Visitor → institute	Pass `visitor_id` + first-touch UTMs in CTA links; save them on lead submit and signup	Voomsales and Vidyayatan forms already post to `admin-core-service/open/v1/audience/lead/submit`	**vacademy.io "Book a demo" opens a Google Calendar link,** so no demo request reaches the backend. Replace it with the lead form. No site captures incoming UTMs yet.
Logins / activity	Read existing session and activity tables	Yes, keyed by institute_id	No product dimension: a login at dash.vacademy.io doesn't say which product (see §2.3).
Product per institute	New `product_entitlements` table (institute, product, state)	`institutes.product` holds a single value (`vacademy` or `vimotion`)	**Biggest gap.** The multi-product institute doesn't exist in the data yet. This is a Phase 0 backend change.
Token usage	Read `ai_usage_event`; map use case → product	Rich data keyed by institute_id	No product column. The use case → product mapping is kept in the portal's Metering catalogue.

Payments	Read admin-core platform billing	Verified Razorpay webhook (payment.*, order.paid, token.*); hourly reconcile for AI subscriptions and credit packs	Only AI plans and credit packs are billed here; core LMS subscription revenue isn't recorded. No `subscription.*` / `payment_link.*` handlers. Don't include per-institute student payments, which are the institutes' own money.
Demo / trial / suspend	Reuse demo provisioning, `institute_trial`, `TrialAccessGuard`	Demo, expiry, login block on expiry	No institute-level suspend or terminate (only per-user deactivate). Add an institute status checked at login.
Telleo / Evalezy usage	Read telephony (admin-core) and open_evaluation (assessment-service)	Both live in the shared backend	Confirm API customers are institutes with the same institute_id.
Cookie consent	Site-wide banner + GA Consent Mode before `visitor_id` is set	Banner on 4 vacademy.io use-case pages only	Missing on every other site.

Access to collect in Phase 0:

- A Google service account, with Viewer access on each GA4 property (including the one inside the GTM container).
- Named owners for the Telleo and Evalezy site repos.
- No new Razorpay keys: the platform credentials are already stored in admin-core.

2.3 Sign-in entry points

- **Vacademy.io, Telleo and Evalezy sign in at `dash.vacademy.io`:** one login and one institute ID for three products.
  - The login can't say which product it's for. Product is decided in two ways:
    - **Arrival:** each site's Login link carries `?product=…&visitor_id=…`, and the dashboard stores it on the session.
    - **Feature use:** calls = Telleo, evaluations = Evalezy, LMS activity = Vacademy.io. Per-product activity and activation use this.
  - Only one app needs instrumenting for these three products. It already sends a `Login Success` event to Amplitude; add a product property.
  - A suspended product can't block login, because the login is shared. Instead, its features are switched off through institute modules.
- **Voomsales signs in at `app.voomsales.com/login`.**
  - In its current code the login is front-end only (it doesn't check credentials), and the CRM screens use a mock institute.

- **Decided:** Voomsales will authenticate through Vacademy auth-service with the same institute IDs.
- Until then, Voomsales contributes website traffic and demo leads only. Accounts, logins, usage and lifecycle actions start working once it's wired.

3. Event tracking plan

All four sites and products send the **same event names with the same parameters**, which is what makes a combined funnel possible.

Event	Fires when	Where
`page_view` / `session_start`	Visitor lands on any page	All marketing sites (GA4 default + tracker)
`cta_login_click`	Login button clicked	All marketing sites (returning-customer metric, not the funnel)
`cta_book_demo_click`	Book a demo clicked	All marketing sites
`demo_form_submit`	Demo lead accepted by the backend lead-submit endpoint	Server side
`cta_trial_click`	Start free trial clicked	All marketing sites
`trial_signup_complete`	Trial account created	Server side
`login_success`	User logs in	Server side, from existing session data
`feature_used`	Key action: course published, lead added, call made, sheet evaluated	Server side
`pricing_view` / `checkout_start`	Pricing page seen / payment link opened	Sites and in-app billing
`payment_success` / `payment_failed`	Platform billing records the result	Server side (admin-core)

vacademy.io already sends `demo_request` to GTM, and Voomsales sends `generate_lead`. Rename both to the standard names, or map them in GA4.

3.1 Parameters on every event

product (`vacademy`, `voomsales`, `telleo`, `evalezy`), **visitor_id**, **institute_id** (once known), **page_path**, **utm_source / utm_medium / utm_campaign**, **device**, **country / state**.

3.2 Linking a website visitor to an institute

The sites are on separate domains (vacademy.io, voomsales.com and the Telleo and Evalezy sites), so they can't share a cookie. Instead:

- **First visit:** after consent, the tracker sets a first-party `visitor_id` cookie and keeps the first-touch UTM values.
- **CTA click:** Login, Book a demo and Start trial links carry `visitor_id`, `product` and the UTMs as URL parameters, to `dash.vacademy.io` or `app.voomsales.com`.
- **Lead submit or signup:** the backend saves `visitor_id` against the new institute ID. From then on, every earlier visit is attributed to that institute.

3.3 GA4 identity

Set GA4 `user_id` to the real user ID, and send `institute_id` as a **user property**. (v1 used the institute ID as the user ID, which would merge every user in an institute into one GA4 "user".) Don't send personal data such as emails or phone numbers to GA4.

3.4 Consent and retention

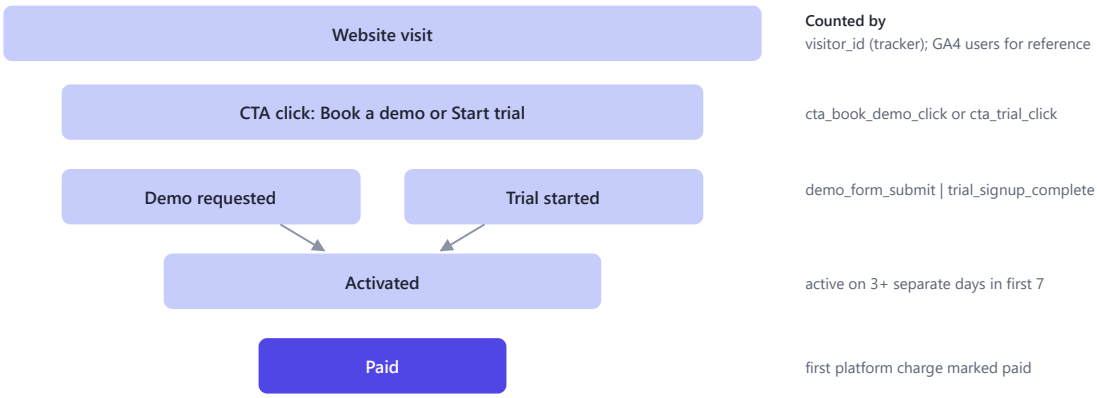
- **Consent banner:** a site-wide banner on all four sites, with GA Consent Mode. `visitor_id` is set only after consent.
- **Retention:** raw web events are kept for 13 months; daily aggregates are kept indefinitely.
- **Visitors who never sign up:** their visitor records are deleted after 13 months.
- **Legal review:** wording and data handling are reviewed against India's DPDP Act 2023 before launch.

4. Dashboard modules

Eight screens in the super admin portal. Every screen has a product filter (All, Vacademy.io, Voomsales, Telleo, Evalezy) and a date range.

Module	What you see	What you can do
Overview	Visitors, demo requests, trial starts, active trials, paid institutes, MRR, today's active users; each with change vs. last period	Click any number to drill into its list
Funnels & traffic	Funnel from visit to paid (below); traffic by source, campaign, landing page and day; returning-customer login clicks	Compare products side by side; export CSV
Accounts list	Every institute: state per product, plan, last login, 7-day active users, tokens used	Search, filter, sort; bulk extend trial
Account 360	One institute: timeline of visits, signup, logins, feature use, tokens, charges, admin actions	All lifecycle actions (§5)
Demos & trials	All demo and trial accounts, days left, activation status, last activity	Create demo, start / extend trial, convert, expire
Revenue	MRR, new vs. churned, failed charges, plan mix, revenue by product, from admin-core platform billing (includes marked-paid charges, excludes waived)	Open the billing record; resend payment link
Alerts	Trials ending in 3 days without activation, failed charges, accounts inactive for 14 days, traffic drops	Mark handled; jump to account
Audit log	Every Command Center action, together with existing module / pricing / plan change history	Filter and export

One funnel, with two branches; filterable by product, date range and UTM source

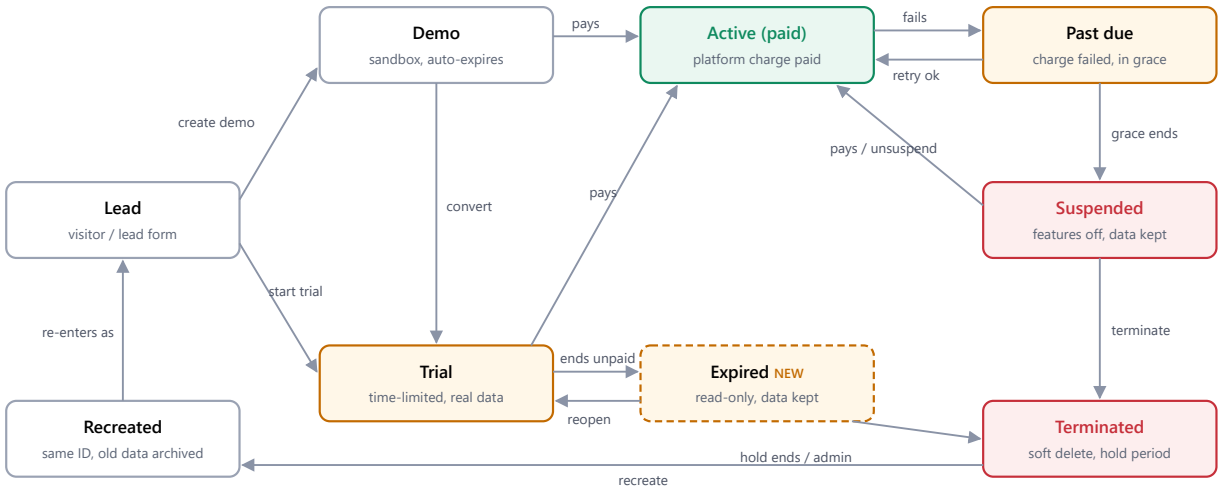


Each stage shows its count and the percentage that moves to the next stage. A demo can also convert straight to paid.

5. Account lifecycle

Each institute has a **state per product**, because an institute can be paid on Vacademy.io while trialling Telleo.

One state machine, per product



5.1 Mapping existing data

- **Lead tags:** `LEAD` → Lead, `FREE_TRIAL` → Trial, `PROD` → Active. `TEST` accounts are excluded from every metric. After migration the tag is derived from the state, so there is one source of truth.
- **Existing demos** in `institute_trial` become Demo state for `vacademy`.

5.2 Actions

Action	Scope	What it does	Safeguard
Create demo	Per product	Existing demo provisioning, from a template per product (sample courses, leads, calls, sheets), assigned to a prospect email	Auto-expires after N days
Start trial	Per product	Enables the product with a trial end date and usage caps	Trial length set per product

Extend trial	Per product	Moves the end date	Reason required and logged
Convert to paid	Per product	Sends a platform billing payment link	Only a paid charge flips the state to Active
Suspend product	Per product	Switches the product's features off through institute modules; login still works for other products	Reversible in one click
Suspend account	Account	Blocks login for the institute (new institute status checked at login)	Named users only; typed confirmation
Terminate	Account	Soft delete; data held, then purged. Lists every product it affects before confirming	Type the institute name; named users only
Recreate	Account	New clean account under the same institute ID. Old data is moved to an archive keyed by archive ID, so it never mixes with new data	Confirmation + audit entry
Reset demo	Per product	Restores the demo template data	Demo accounts only

6. Data model

New tables go in the `analytics` schema; existing tables are read, not changed. The one exception is the Phase 0 entitlements change.

Table	Key columns	Purpose
product_entitlements prerequisite	institute_id, product, state, plan, trial_ends_at, created_by	Current state per product; replaces the single `institutes.product` value
lifecycle_events	institute_id, product, from_state, to_state, reason, actor, at	History behind the state machine; feeds the Audit log
visitors	visitor_id, first_utm_*, first_seen_at, institute_id	Visitor-to-institute link
web_events	visitor_id, product, event, page_path, utm_*, at	First-party funnel events
ga4_daily	date, product, source, medium, campaign, sessions, users, key events	Synced GA4 aggregates
usage_daily	date, institute_id, product, active_users, sessions, minutes, tokens	Nightly rollup from existing session, activity and `ai_usage_event` data
use_case_product_map	use_case, product	Gives token usage a product

Dropped from v1:

- `subscriptions / payments`: read from admin-core platform billing instead.
- A separate `admin_audit_log`: replaced by `lifecycle_events` plus the existing change history.
- `product_events` for logins: logins already exist in the session tables.

7. Metrics glossary

One definition for every number, used on every screen.

Metric	Definition
Visitors	Unique `visitor_id`s on a product site in the period. GA4 users are shown alongside for reference only; they will differ because of ad blockers and consent.
CTA click rate	Visitors who clicked Book a demo or Start trial, divided by visitors
Returning-customer clicks	Visitors who clicked Login (outside the funnel)
Demo requests	`demo_form_submit` events accepted by the backend
Trial starts	`trial_signup_complete` events
Activated trial	Trial institute with activity in that product on 3 or more separate days in its first 7 days
Trial-to-paid rate	Trials whose product reaches Active within 14 days **after** the trial end date, divided by trials that ended in the period
DAU / WAU / MAU	Users with a session in the last 1 / 7 / 30 days. Per product: users with feature use in that product.
Avg session length	Mean minutes from login to last activity, per session (from `user_session`)
Tokens per institute	From `ai_usage_event`, per product (via use case mapping), per period
MRR	Sum of active monthly subscription values in platform billing (annual plans / 12). Includes manually marked-paid charges; excludes waived charges.
Churn	Products moving from Active to Terminated, or to Suspended past the grace period, divided by Active at the start of the period. Past-due accounts still in grace don't count.

Thresholds such as "3 days in 7" are starting points; tune them after a month of data.

8. Rollout

Visibility ships first. Revenue moves up to Phase 1 because its data already exists.

About 16 weeks for one small team (suggested)



Dependencies

- **Backend (Phase 0):** the `product_entitlements` table, and an institute status checked at login for account suspend.
- **GA4 access:** property IDs and Viewer access for the service account, including the GA4 property inside the vacademy.io GTM container.

- **Site work on all four sites:** tracker snippet, standard CTA event names, CTA links that carry `visitor_id` / `product` /UTMs, and a consent banner.
- **Owners:** named owners for the Telleo and Evalezy sites.
- **vacademy.io demo form:** "Book a demo" posts to the backend lead endpoint instead of only opening Google Calendar.
- **Voomsales:** the login uses Vacademy auth-service and real institute IDs instead of the mock institute.
- **Revenue coverage:** core LMS subscriptions are billed through platform billing. Until they are, Revenue covers AI plans and credit packs only.
- **Demo templates:** one template institute per product, with realistic sample data (Phase 2).

Open questions

- Trial length and usage caps per product (for example, 14 days and a token cap on Telleo)?
- How long is an Expired trial kept before it can be terminated, and how long is terminated data held before purge? (30 days is the default.)
- Should prospects be able to self-serve a demo from the website, or is demo creation PM-only?
- Which plans and prices go into platform billing first, and are any annual?
- Which usernames may run Suspend account, Terminate and Recreate?
- Are Telleo and Evalezy API customers always institutes with a Vacademy institute ID, or can an API key belong to someone else?
- Ingestion host: community-service (proposed) or Cloudflare cron triggers?
